

Experiment 15

Demonstrate how to work with Git branches and resolve merge conflicts when collaborating with others.

Tasks:

1. Clone a shared repository to your local machine.
2. Create a new branch and switch to it.
3. Make changes to a file (e.g., update a README or modify
4. code in a specific function).
5. Commit your changes.
6. Push the changes to the remote repository.
7. Before merging, pull the latest changes from the main branch.
8. Switch back to your feature branch.
9. Merge the main branch into your feature branch.
10. If there are conflicts, resolve them manually by editing the
11. conflicting files. After resolving conflicts, mark the conflicts
12. as resolved.
13. Commit the resolved merge.
14. Push your feature branch with the merged changes to GitHub
15. and create a pull request.
16. Submit a summary of the steps you performed, the conflicts
17. you encountered, and how you resolved them.

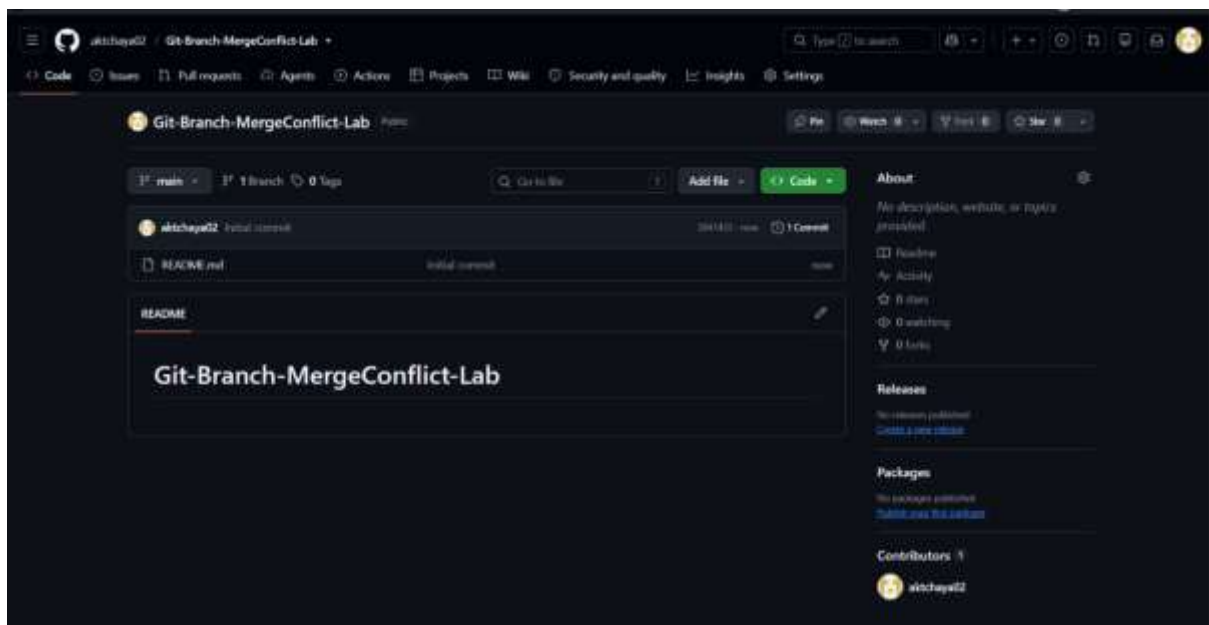
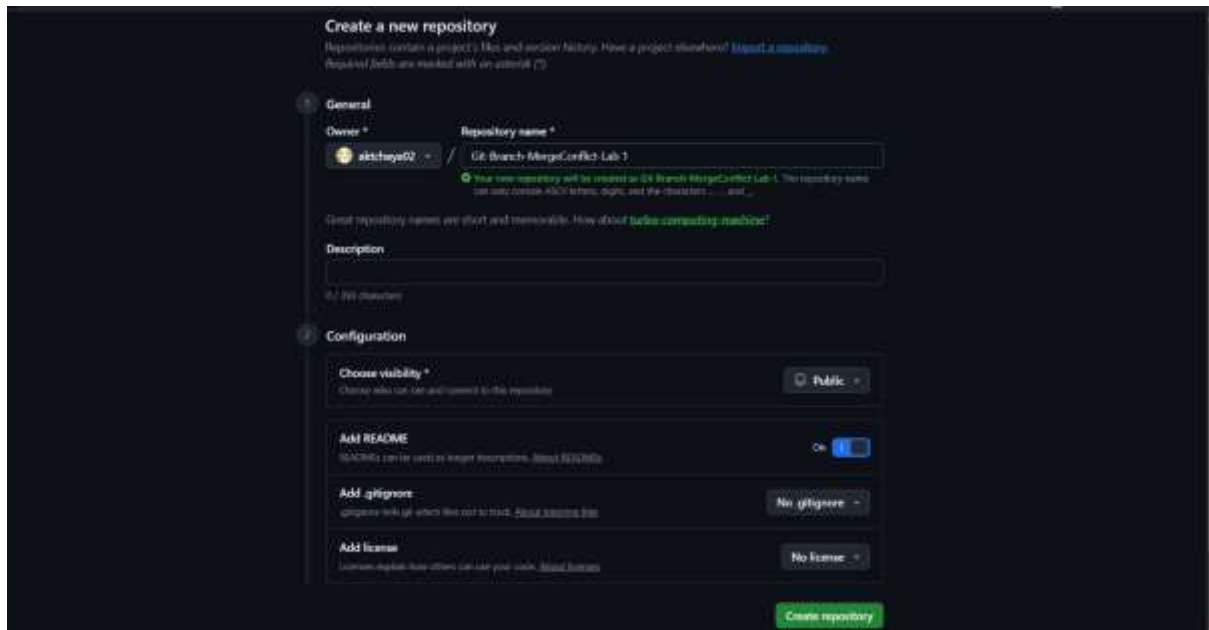
Step 1: Create a New GitHub Repository

1. Log in to your GitHub account.
2. Click the + icon and select **New repository**.
3. Enter the repository name:

Git-Branch-MergeConflict-Lab

4. Select **Public**.
5. Check **Add a README file**.
6. Click **Create repository**.

This creates a new GitHub repository with a default README.md file.



Step 2: Clone the Repository

Cloning means downloading a copy of the GitHub repository to your local computer.

1. Copy the repository HTTPS URL.
2. Open **Command Prompt**.
3. Run:

```
git clone https://github.com/YourUsername/Git-Branch-MergeConflict-Lab.git
```

4. Move into the repository folder:

cd Git-Branch-MergeConflict-Lab

```
C:\Users\Hp>git clone https://github.com/aktchaya02/Git-Branch-MergeConflict-Lab.git
Cloning into 'Git-Branch-MergeConflict-Lab'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

C:\Users\Hp>cd Git-Branch-MergeConflict-Lab
C:\Users\Hp\Git-Branch-MergeConflict-Lab>
```

Step 3: Create a New Branch

1. Instead of working directly on the main branch, create a separate branch.
2. Run:

```
git checkout -b feature-update
```
3. Verify the branches:

```
git branch
```
4. The current branch will have an asterisk (*).

```
C:\Users\Hp\Git-Branch-MergeConflict-Lab>git checkout -b feature-update
Switched to a new branch 'feature-update'

C:\Users\Hp\Git-Branch-MergeConflict-Lab>git branch
* feature-update
main
```

Step 4: Modify the README File in the Feature Branch

1. Open the repository folder in **Visual Studio Code**.
2. Open **README.md**.
3. Replace or update the contents with:

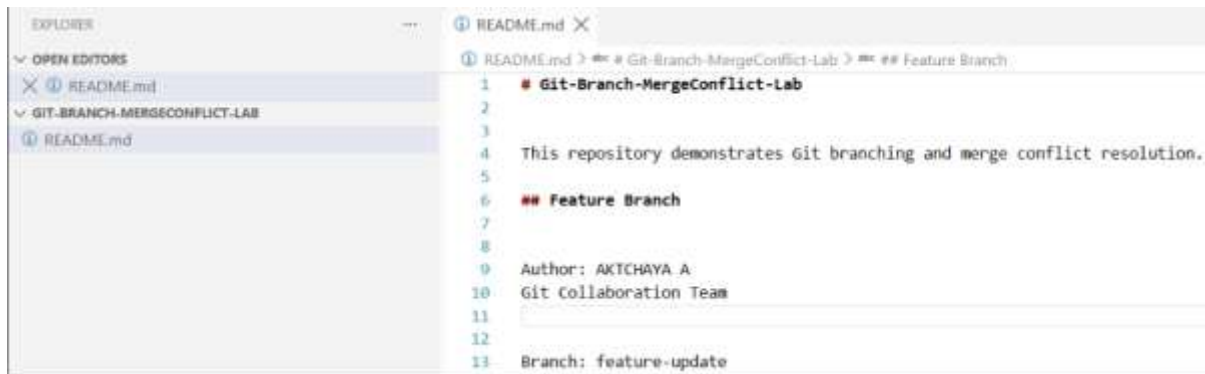
```
# Git-Branch-MergeConflict-Lab

This repository demonstrates Git branching and merge conflict resolution.

## Feature Branch

Author: AKTCHAYA A

Branch: feature-update
```
4. Save the file using **Ctrl + S**.



Step 5: Commit and Push the Feature Branch

1. Check the status:

```
git status
```

2. Stage the file:

```
git add README.md
```

3. Commit the changes:

```
git commit -m "Updated README from feature branch"
```

4. Push the branch:

```
git push origin feature-update
```

```
C:\Users\Hp\Git-Branch-MergeConflict-Lab>git add README.md

C:\Users\Hp\Git-Branch-MergeConflict-Lab>git commit -m "Updated README from
feature branch"
[feature-update 1c68cd5] Updated README from feature branch
1 file changed, 9 insertions(+), 1 deletion(-)

C:\Users\Hp\Git-Branch-MergeConflict-Lab>git push origin feature-update
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 405 bytes | 405.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-update' on GitHub by visiting:
remote:   https://github.com/aktchaya02/Git-Branch-MergeConflict-Lab/pull
/new/feature-update
remote:
To https://github.com/aktchaya02/Git-Branch-MergeConflict-Lab.git
 * [new branch]      feature-update -> feature-update
```

Step 6: Update the Main Branch

1. Switch to the main branch:

```
git checkout main
```

2. Open **README.md** again.
3. Modify the **same line** to create a merge conflict.
4. For example, change:

Author: AKTCHAYA A

to:

Author: Git Collaboration Team

Save the file.

5. Stage the file:

```
git add README.md
```

6. Commit the changes:

```
git commit -m "Updated README in main branch"
```

7. Push the changes:

```
git push origin main
```

```
C:\Users\Hp\Git-Branch-MergeConflict-Lab>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

C:\Users\Hp\Git-Branch-MergeConflict-Lab>git add README.md

C:\Users\Hp\Git-Branch-MergeConflict-Lab>git commit -m "Updated README in main branch"
[main 01bc9a0] Updated README in main branch
 1 file changed, 10 insertions(+), 1 deletion(-)

C:\Users\Hp\Git-Branch-MergeConflict-Lab>git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 401 bytes | 401.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/aktchaya02/Git-Branch-MergeConflict-Lab.git
 3841435..01bc9a0  main -> main
```

Step 7: Merge the Main Branch into the Feature Branch

1. Switch back to the feature branch:

```
git checkout feature-update
```

2. Merge the latest changes from main:

`git merge main`

3. Since the same line was modified in both branches, Git displays a merge conflict.

4. Example:

CONFLICT (content): Merge conflict in README.md

Automatic merge failed; fix conflicts and then commit the result.

```
C:\Users\Hp\Git-Branch-MergeConflict-Lab>git checkout feature-update
Switched to branch 'feature-update'

C:\Users\Hp\Git-Branch-MergeConflict-Lab>git pull origin main
From https://github.com/aktchaya02/Git-Branch-MergeConflict-Lab
 * branch          main          -> FETCH_HEAD
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.

C:\Users\Hp\Git-Branch-MergeConflict-Lab>git merge main
error: Merging is not possible because you have unmerged files.
hint: Fix them up in the work tree, and then use 'git add/rm <file>'
hint: as appropriate to mark resolution and make a commit.
fatal: Exiting because of an unresolved conflict.
```

Step 8: Resolve the Merge Conflict

1. Open Visual Studio
2. Open README.md.
3. Git highlights the conflict using markers like:

<<<<<<< HEAD

Author: AKTCHAYA A

=====

Author: Git Collaboration Team

>>>>>>> main

Remove the conflict markers

4. Keep the final version:

Git-Branch-MergeConflict-Lab

This repository demonstrates Git branching and merge conflict resolution.

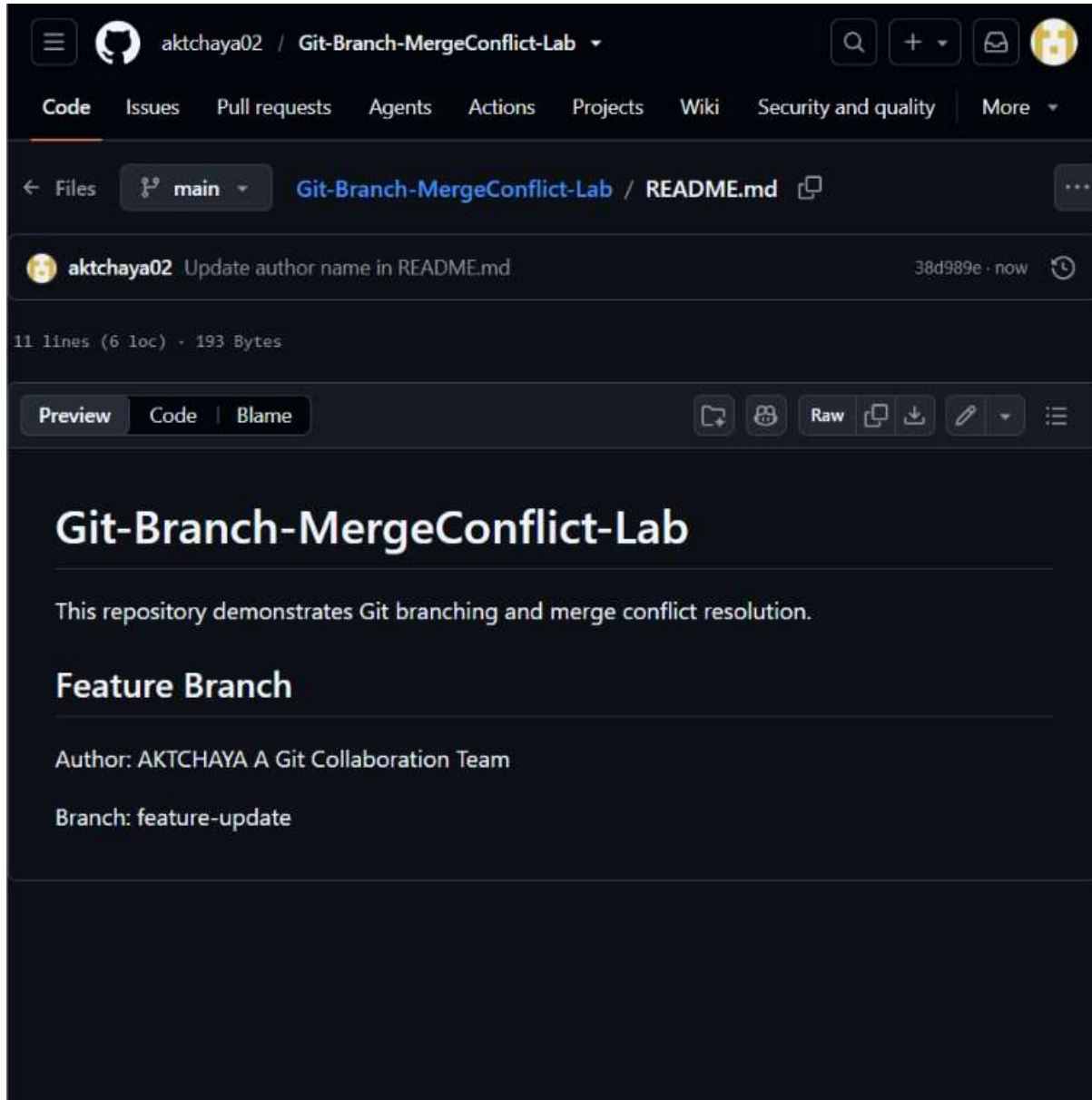
Feature Branch

Author: AKTCHAYA A

Git Collaboration Team

Branch: feature-update

5. Save the file.



Step 9: Commit the Resolved Conflict

1. Stage the resolved file:

```
git add README.md
```
2. Commit the resolution:

```
git commit -m "Resolved merge conflict"
```

```
C:\Users\Hp\Git-Branch-MergeConflict-Lab>git add README.md

C:\Users\Hp\Git-Branch-MergeConflict-Lab>git commit -m "Resolved merge conflict
[feature-update ec5f9ed] Resolved merge conflict
```

Step 10: Push the Updated Feature Branch

Push the resolved branch:

```
git push origin feature-update
```

```
C:\Users\Hp\Git-Branch-MergeConflict-Lab>git push origin feature-update
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 756 bytes | 756.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/aktchaya02/Git-Branch-MergeConflict-Lab.git
  1c68cd5..ec5f9ed  feature-update -> feature-update
```

Step 11: Create a Pull Request

1. Open the GitHub repository.
2. Click **Compare & pull request**.
3. Enter:

Title

Resolved merge conflict and updated README

Description

This pull request includes the following changes:

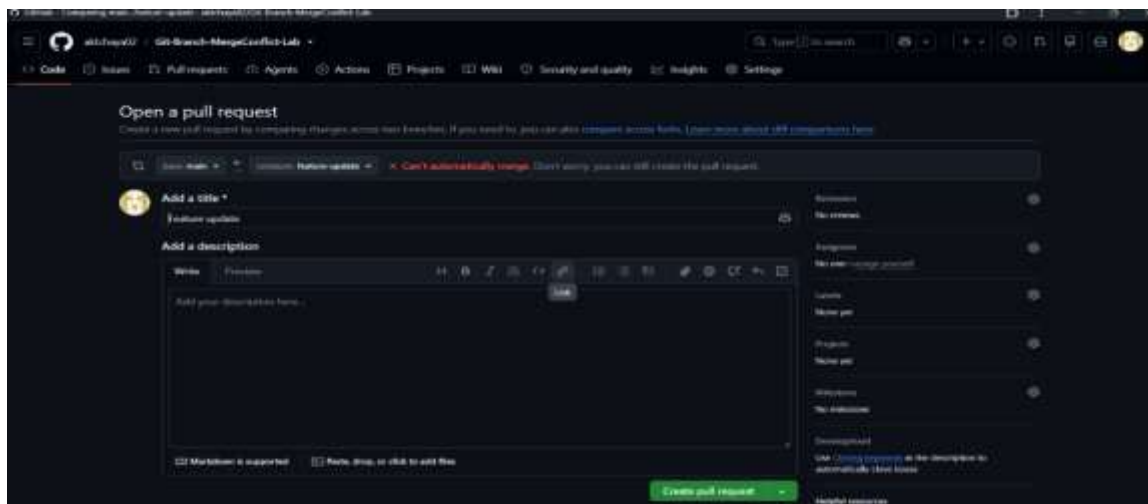
Updated the README.md file in the feature-update branch.

Merged the latest changes from the main branch.

Resolved the merge conflict manually.

Verified the changes after resolving the conflict.

4. Click **Create Pull Request**.



Step 12: Merge the Pull Request

1. Open the **Conversation** tab.
2. Click **Merge Pull Request**.
3. Click **Confirm Merge**.
4. Optionally, click **Delete Branch**.
5. GitHub displays:

Pull request successfully merged and closed

